CHAPTER 6

LANDFORMS AND THEIR EVOLUTION

CHAPTER ESSENCE AT A GLANCE:
Examine the dynamic erosional and depositional landforms created by running water, groundwater (karst), glaciers, wind (aeolian), and waves along coastlines, tracing the geomorphic evolution of landscapes.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Fluvial Landform Dynamics:** Running water erodes vertically in its youthful stage forming gorges, canyons, and waterfalls; laterally in mature stages forming meanders and ox-bow lakes; and deposits alluvium in old age forming floodplains and deltas.

• **Groundwater Karst Topography:** In soluble limestone terrains, carbonic acid dissolves rocks along joint planes forming sinkholes, swallow holes, lapies, and uvalas; precipitation of calcium carbonate inside caves yields stalactites, stalagmites, and pillars.

• **Glacial Erosion and Glaciation:** High-altitude glaciers pluck and abrade bedrock into amphitheatre-like cirques, horns, aretes, and U-shaped valleys; melting glaciers deposit unstratified till in moraines (lateral, medial, terminal), drumlins, and eskers.

• **Aeolian and Coastal Geomorphology:** Wind deflates desert floors forming mushroom rocks, yardangs, and barchan dunes; ocean waves erode sea cliffs and wave-cut platforms, depositing sand into spits, bars, barriers, and sandy beaches.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Stream Competence (Maximum Mass)} \propto v^{6} \rightarrow \text{Sixth Power Velocity Law}$$
Hydraulic Competence Power Law of Bedload Transport

$$\text{Fluvial Profile Evolution} \rightarrow \text{Youth (V-Valley)} \rightarrow \text{Mature (Meanders)} \rightarrow \text{Old (Peneplain)}$$
Davisian Cycle of Fluvial Landscape Evolution

$$\text{Karst Dissolution Sequence} \Longleftrightarrow \text{Sinkholes} \rightarrow \text{Doline} \rightarrow \text{Uvala} \rightarrow \text{Polje}$$
Progressive Morphological Coalescence of Karst Depressions

$$y = a \cdot x^{b} \quad (b \approx 2) \rightarrow \text{Parabolic Glacial U-Shaped Trough}$$
Mathematical Geometry of Glacially Carved Valleys

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Stalactites grow downwards from the ceiling of a limestone cave, while stalagmites grow upwards from the cave floor; when they merge, they form a continuous cave pillar.

• **Alert:** Barchans are crescent-shaped sand dunes with horns pointing downwind (in the direction of the prevailing wind), whereas parabolic dunes have their horns pointing upwind.

• **Alert:** Hanging valleys are created when tributary glaciers have much less ice volume and erosive power than the trunk glacier, leaving their valley floors stranded high above the main U-shaped valley floor.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between a Gorge and a Canyon with suitable examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Gorge:** A deep, narrow valley with very steep, near-vertical rocky walls, having equal width at top and bottom (e.g. Indus Gorge in Kashmir). **[1 Mark]**

• **Canyon:** A deep valley characterized by stepped slopes with wider top than bottom, typically carved through horizontal sedimentary rock strata in arid climates (e.g. Grand Canyon of the Colorado River). **[1 Mark]**

2. How do stalactites, stalagmites, and cave pillars form in limestone caves? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Stalactites and Stalagmites:** Stalactites are icicle-like calcium carbonate deposits hanging down from cave roofs formed by dripping water; stalagmites grow upward from the cave floor where drops fall. **[1 Mark]**

• **Cave Pillars:** Formed when a descending stalactite and an ascending stalagmite join together over centuries to form a continuous column. **[1 Mark]**

3. Explain the formation of Meanders and Ox-bow lakes by running water in its mature stage. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Meander Formation:** In the mature stage, lateral erosion dominates; small irregularities in stream gradient cause the river to oscillate, eroding the outer bank (concave bank) and depositing on the inner bank (convex slip-off bank). **[1.5 Marks]**

• **Ox-bow Lake Formation:** Continuous lateral erosion narrows the meander neck until floodwaters cut straight through the neck, abandoning the curved loop as an isolated crescentic ox-bow lake. **[1.5 Marks]**

4. Describe the erosional glacial features known as Cirques, Aretes, and Horns. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Cirque:** A deep, amphitheatre-shaped armchair depression with steep vertical headwalls carved by glacial plucking at mountain heads; when filled with water it forms a tarn lake. **[1 Mark]**

• **Arete:** A sharp, knife-edged, narrow rocky ridge separating two adjacent cirques. **[1 Mark]**

• **Horn:** A high, pyramid-shaped steep peak formed when three or more cirques erode headward towards a central summit (e.g. Matterhorn in the Alps). **[1 Mark]**

5. What are Barchans and Seifs? Describe the mechanisms of wind erosion and deposition in arid desert environments. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Barchans:** Crescent-shaped sand dunes with convex slopes facing upwind and two horns pointing downwind in the direction of prevailing wind. **[1 Mark]**

• **Seifs:** Long, narrow longitudinal sand ridges running parallel to the prevailing wind direction with a single wing. **[1 Mark]**

• **Wind Erosion Mechanisms:** Operates via Deflation (lifting and blowing away loose sand particles creating deflation hollows) and Abrasion (sand-blasting bedrock to create mushroom rocks and yardangs). **[1 Mark]**

• **Wind Deposition:** Occurs when wind velocity drops or obstacles impede movement, depositing sorted fine silt (loess) or coarse sand dunes. **[1 Mark]**

6. Discuss the erosional and depositional landforms created by waves along rocky and sandy coastlines. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Erosional Coastal Landforms:** High wave energy attacks headlands, cutting horizontal wave-cut notches that collapse into steep Sea Cliffs; retreating cliffs leave flat Wave-cut Platforms; refraction creates Sea Caves, Arches, and isolated Stacks. **[2 Marks]**

• **Depositional Coastal Landforms:** Wave swash and backwash deposit sand and shingles forming Beaches; longshore drift extends sandy ridges into the sea as Spits and across bays as Bars, which enclose coastal Lagoons (e.g. Chilika Lake). **[2 Marks]**